

Helena Montoya Calero

PhD Candidate · Bocconi University · ION Management Science Lab

Milan, Italy helena.montoya@phd.unibocconi.it helenamontoyacalero.com LinkedIn: [helenamontoya-calero](#)

EDUCATION

Bocconi University

2023 — present

PhD in Management · Focus: Decision-Making under Uncertainty · Milan, Italy

Advisors: Alfonso Gambardella and Arnaldo Camuffo · ION Management Science Lab

Université Catholique de Louvain

2019 — 2021

Master 120 in Economics, Research track · Louvain, Belgium

Thesis: Measuring equality of opportunity based on individual responsibility — comparison Belgium and Colombia. Advisor: François Maniquet.

EAFIT University

2014 — 2018

Economics Bachelor · Medellín, Colombia

WORKING PAPERS

Measuring Scientific Decision-Making in Organizations

with Arnaldo Camuffo and Thomaz Teodorovicz

This research note introduces a 24-item survey instrument for measuring scientific decision-making (SDM) in organizations, reports diagnostic and within-study consistency checks, and applies it to a field sample of 3,178 respondents in 54 firms (analytic $N = 2,745$). Scientific decision-making is increasingly treated as a managerial resource, yet scalable tools to observe reported decision practices inside firms remain scarce. Variation is concentrated within firms: the overall SDM intraclass correlation is 0.018, implying that only 1.8% of variance is attributable to firm membership. Adaptive Learning, the dimension capturing whether managers revise beliefs in response to new evidence, shows the clearest top-tier contrast: Tier 1 managers score higher than frontline managers (unconditional $\Delta = .17$; with education, tenure, and decision-area controls, $b = .14-.16$, $p < .001$, without and with firm fixed effects), while Causal Reasoning and Evidence Evaluation do not show comparable top-tier patterns. SDM is also positively associated with performance on a short within-study decision vignette ($r = .19$), with a 9.3-percentage-point top-versus-bottom-quintile gap on vignette accuracy. The contribution of this note is a survey module and a compact within-study benchmark.

$N = 3,178$ · 54 firms

Heterogeneous Effects of Theory-Based Entrepreneurship Training

with Arnaldo Camuffo and Alfonso Gambardella

Examines heterogeneous treatment effects of theory-based training programs across five RCTs (Italy, Colombia, Netherlands, Spain, UK) using Random Forest and causal inference methods.

5 RCTs · In progress

Extracting Interpretable Models from Large Language Models for Organizational Data

with Said Kazemi

Compares classical econometric approaches (LASSO) with LLM-based methods for predicting ordinal organizational variables. Explores linear probing, prompt-ensemble, and linear distillation techniques.

LLMs · In progress

EXPERIENCE

ION Management Science Lab, Bocconi University

Jun 2023 — present

Research Assistant · Milan, Italy

Decision-Making under Uncertainty, Management Science, field experiments. PIs: Alfonso Gambardella and Arnaldo Camuffo.

Fundación Éxito

Sep 2021 — May 2022

Research Associate · Medellín, Colombia

Research on chronic malnutrition and treatment programs for mothers and children across Colombia. Supported design and implementation of mother-assistance projects, including regular data collection and program monitoring.

Innovations for Poverty Action (IPA)

Apr 2020 — Jun 2021

Research Associate · Medellín, Colombia

PI: Christopher Blattman (University of Chicago). Projects on criminal governance (Gang Rule) and youth decision-making in contexts of violence (Recruitment).

EAFIT University

2016 — 2020

Research Assistant · Medellín, Colombia

Research on labor discrimination in migration, citizen security, and cooperative experiment design. Collaborated with PIs at EAFIT, Los Andes, and the World Bank.

CONFERENCES & PRESENTATIONS

Utah Winter Strategy Seminar

Jan 2025

Deer Valley, Utah

Theory-Based View: A Scientific Approach to Value Creation

2nd Annual Bayesian Entrepreneurship Conference

2025

Bocconi University · Milan, Italy

SKILLS

Programming & data: Python, R, STATA, LaTeX, GIS, API integration, LLMs

Methods: Causal inference, field experiments, machine learning, NLP

Languages: Spanish (native) · English (TOEFL 97)